Anomaly Detection in Ship Engine Data

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Anomaly Detection in Ship Engine Data

Abstract

This report presents an anomaly-detection study conducted on a dataset containing six key operational parameters from a ship engine. The analysis includes exploratory data assessment, data cleansing, distribution analysis, correlation evaluation, and both statistical and machine-learning-based anomaly detection. A univariate IQR approach was tested but found unsuitable for this dataset, leading to the adoption of the Isolation Forest algorithm for multivariate anomaly detection. The findings highlight patterns of abnormal engine behaviour that may indicate mechanical stress, lubrication issues, or cooling inefficiencies.

1. Introduction

Ship engines operate under fluctuating mechanical and thermal conditions, making continuous monitoring essential for safety and reliability. Sensor data provides valuable insight into engine performance, but identifying abnormal patterns manually can be challenging. This project applies a structured analytical workflow to detect anomalies within a dataset of six engine-related features. The goal is to identify unusual operating states that may signal emerging faults or inefficiencies.

2. Dataset Description

The dataset consists of six numerical features representing core engine functions:

- Engine rpm
- Lubrication oil pressure
- Fuel pressure
- Coolant pressure
- Lubrication oil temperature
- Coolant temperature

These variables collectively describe the engine’s mechanical load, lubrication system behaviour, fuel system performance, and thermal conditions.

3. Exploratory Data Analysis

3.1 Structural Review

An initial inspection confirmed that all six features were numerical and that the dataset contained a large number of observations typical of continuous sensor logging. Summary statistics showed expected operating ranges, with rpm and temperature-related features exhibiting broader variability than pressure-based measurements.

3.2 Visual Exploration

Histograms and density plots were used to examine the distribution of each feature. These visualisations revealed skewness in several variables and suggested the presence of extreme values. A correlation of heatmap was also generated to assess relationships between features. The heatmap showed minimal correlation across the dataset, indicating that anomalies may arise from complex interactions rather than simple linear dependencies.

4. Data Cleansing

4.1 Missing Values

A check for null values showed that the dataset was largely complete. Any missing entries were addressed to ensure consistency before further analysis.

4.2 Duplicate Records

Duplicate rows were identified and removed to prevent repeated sensor readings from influencing the analysis. This step ensured that each observation represented a unique engine state.

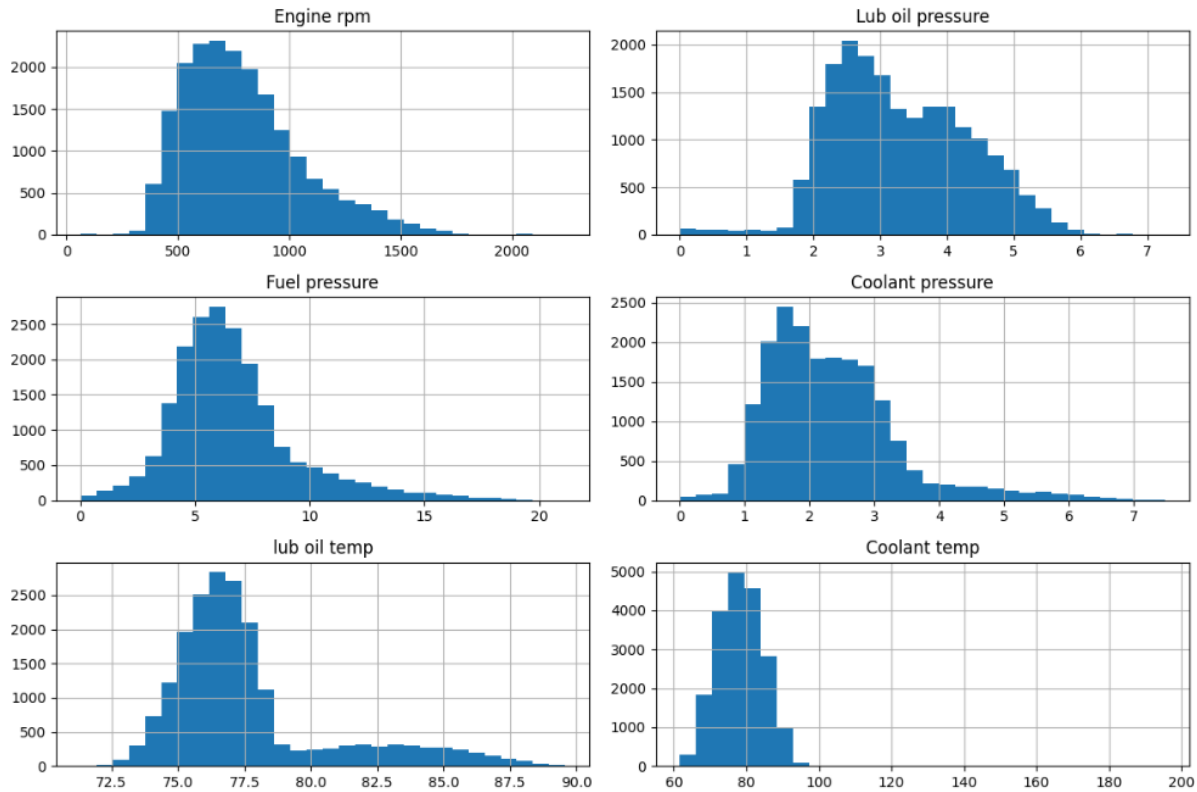
4.3 Outlier Review

Extreme values were examined to distinguish between genuine anomalies and sensor errors. Only clear logging errors were removed, preserving meaningful anomalies for modelling.

5. Distribution Analysis

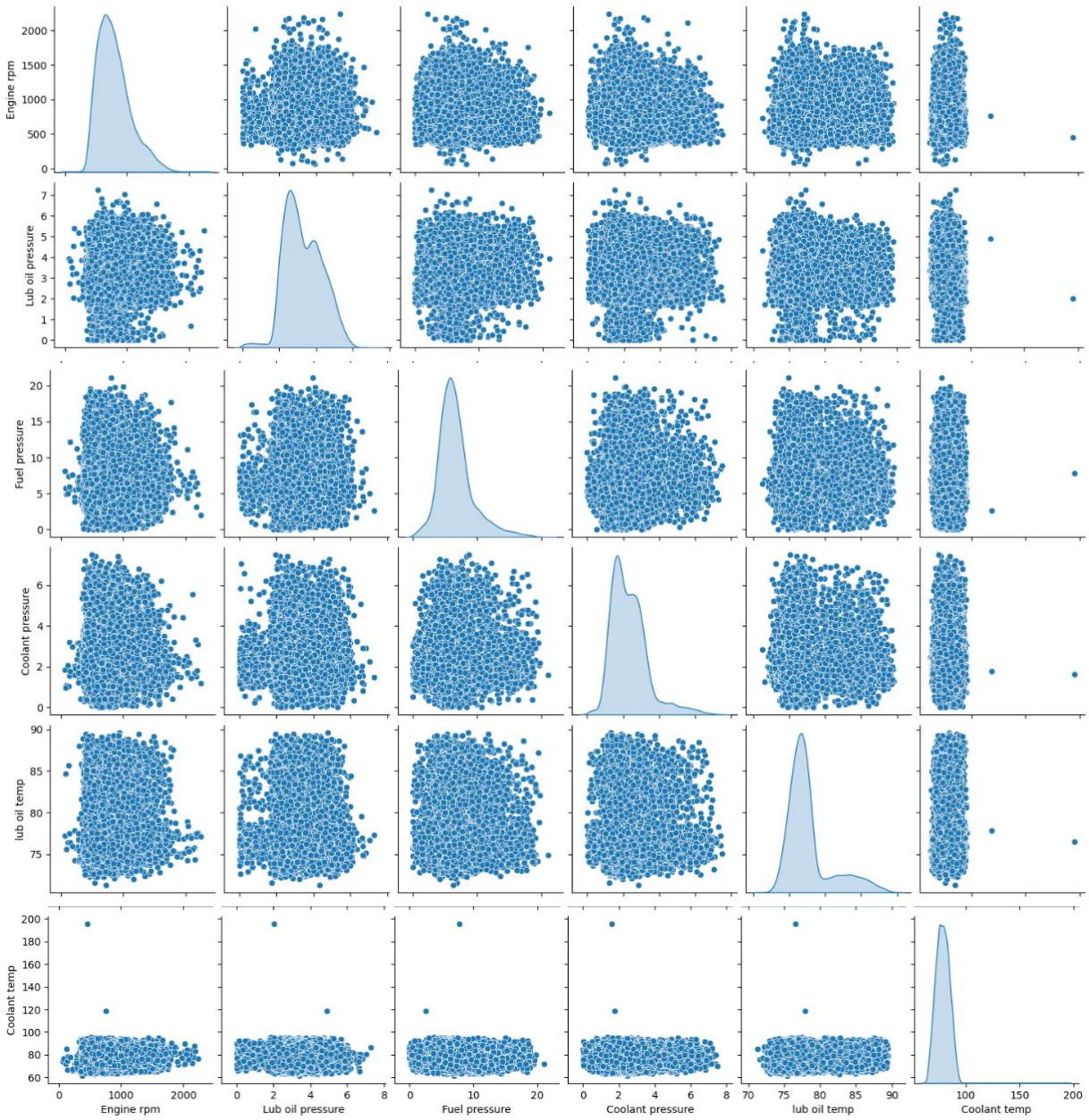
Understanding the statistical behaviour of each feature is essential for identifying unusual patterns. Key observations included:

The features show skewness and are not exactly normally distributed. Below is the histogram chart for each of the attributes.



6. Pair Plot Analysis

A pair plots was generated to visualise relationships between all feature combinations. The scatter plots revealed clusters corresponding to different engine states and highlighted isolated points that may represent unusual or rare operating conditions. The pair plot also supported the earlier finding that the features were not strongly correlated.



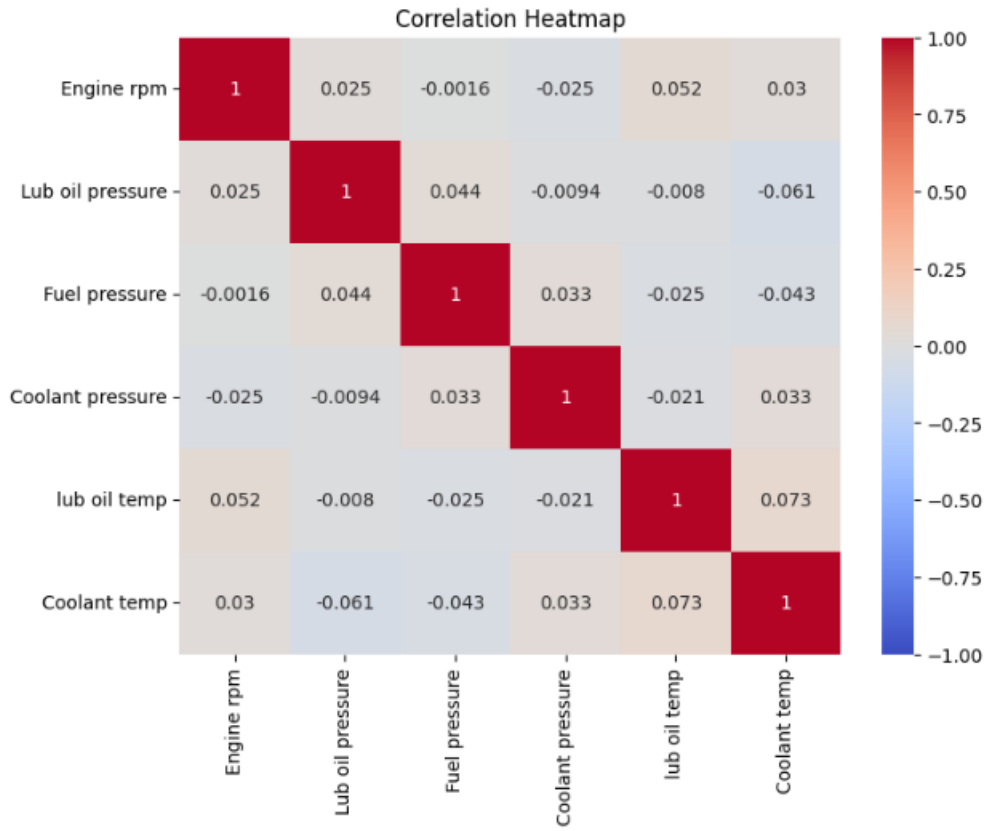
7. Box Plot Analysis

Box plots were created for each feature to identify potential outliers. These visualisations highlighted:

- High-rpm readings associated with heavy engine load
- Low lubrication oil pressure values that may indicate lubrication issues
- Elevated temperatures that could reflect overheating or friction
- Variations in fuel pressure that may point to injection irregularities

8. Correlation Assessment

A correlation matrix was computed to evaluate relationships between the six features. The analysis showed that the variables were largely uncorrelated, suggesting that anomalies are likely to arise from multivariate interactions rather than simple pairwise relationships. This reinforced the need for a multivariate anomaly-detection model.



- Deep red indicates strong positive correlation.
- Deep blue indicates strong negative correlation.
- White or pale colours indicate little or no correlation.

9. Statistical Anomaly Detection Using IQR

A univariate anomaly-detection method based on the Interquartile Range (IQR) was applied to each feature. Although this approach successfully flagged extreme values, it was not suitable for this dataset because:

- Engine behaviour is multivariate
- Many anomalies arise from combinations of features
- Some variables naturally have wide ranges, making IQR thresholds unreliable

```
df_iqr.head()

#iqr_outlier = {}

for col in df_iqr.columns:
    Q1 = df_iqr[col].quantile(0.25)
    Q3 = df_iqr[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    mask = df_iqr[(df_iqr[col] < lower) | (df_iqr[col] > upper)]

mask

#iqr_outliers[col] = df_iqr[mask].index.tolist()
```

	Engine rpm	Lub oil pressure	Fuel pressure	Coolant pressure	lub oil temp	Coolant temp
3242	760	4.908028	2.594570	1.747730	77.873482	118.371957
10663	455	2.010802	7.806127	1.619575	76.519385	195.527912

10. Isolation Forest for Multivariate Anomaly Detection

10.1 Model Rationale

Isolation Forest was selected due to its ability to handle numerical data efficiently, its independence from distributional assumptions, and its effectiveness in isolating unusual points through recursive partitioning.

10.2 Model Configuration

The model was trained on scaled versions of the selected features to ensure equal weighting. Key parameters included:

- 300 estimators
- Contamination rate of 0.05
- Fixed random seed for reproducibility

10.3 Results

The model produced two outputs:

- **Anomaly flag** (1 for normal, –1 for anomaly)
- **Anomaly score**, where lower values indicate stronger anomalies

Approximately 5% of the dataset was classified as anomalous. These cases involved unusual combinations of rpm, pressures, and temperatures. Some anomalies occurred at high rpm, others at low lubrication oil pressure, and several at elevated temperatures. These patterns may correspond to mechanical stress, lubrication problems, or cooling inefficiencies.

Model with all the features taken into consideration:

	Engine rpm	Lub oil pressure	Fuel pressure	Coolant pressure	lub oil temp	Coolant temp	anomaly_flag	anomaly_score
0	682	2.391656	4.617196	2.848982	76.272417	69.884609	1	0.116000
1	605	5.466877	6.424361	5.727520	73.222679	74.907314	-1	-0.024061
2	658	3.434232	3.680896	1.678708	88.089916	78.704806	1	0.022535
3	749	2.094656	7.120927	1.639670	77.661625	82.386700	1	0.134200
4	676	3.538228	5.956472	3.225336	75.226352	67.153220	1	0.091356
5	589	2.933888	6.358464	2.461478	77.534616	75.403745	1	0.153663
6	885	2.034509	15.499680	1.522790	78.959645	80.817985	1	0.025723
7	576	5.495972	13.114658	1.251058	78.091390	71.934674	1	0.017270
8	535	3.159440	8.676361	1.265139	82.408722	80.752747	1	0.082475
9	806	3.946885	21.138326	1.572569	74.923316	72.112692	1	0.011138

```
anomalies = df_iso[df_iso["anomaly_flag"] == -1]
anomalies.count()
anomalies.head(10)
```

	Engine rpm	Lub oil pressure	Fuel pressure	Coolant pressure	lub oil temp	Coolant temp	anomaly_flag	anomaly_score
1	605	5.466877	6.424361	5.727520	73.222679	74.907314	-1	-0.024061
17	1312	5.963439	7.563578	5.931953	77.738876	88.661972	-1	-0.072880
35	1374	2.543575	8.641020	1.541131	82.268575	66.810303	-1	-0.016524
113	1495	3.276230	3.714396	2.495264	85.053243	75.743004	-1	-0.006183
131	830	3.231908	13.479172	2.568096	87.480362	73.428213	-1	-0.017460
144	1762	1.697487	4.327845	2.523372	86.750349	82.393211	-1	-0.064417
149	1193	3.669258	7.845989	1.229372	85.322621	67.009104	-1	-0.017045
157	534	5.922045	18.044686	5.946436	77.292791	74.734372	-1	-0.099925
160	1546	3.903522	6.205256	5.554939	78.825641	73.677703	-1	-0.013974
176	626	4.632698	9.455324	0.458069	86.205094	79.933331	-1	-0.015227

Model with three features taken into consideration:

```
anomalies_1 = df_iso_1[df_iso_1["anomaly_flag"] == -1]
#anomalies_1.count()
anomalies_1.head(10)
```

	Engine rpm	Coolant pressure	lub oil temp	anomaly_flag	anomaly_score
1	605	5.727520	73.222679	-1	-0.048858
2	658	1.678708	88.089916	-1	-0.007354
13	978	2.829643	88.038301	-1	-0.020653
16	1023	3.409817	86.485544	-1	-0.016512
17	1312	5.931953	77.738876	-1	-0.060987
39	1026	5.788750	76.053238	-1	-0.008876
73	1438	2.327817	84.176073	-1	-0.006244
98	617	6.197552	76.404974	-1	-0.021416
113	1495	2.495264	85.053243	-1	-0.034480
122	1454	1.260121	83.372256	-1	-0.010556

11. Conclusion

This project applied a structured analytical workflow to detect anomalies in ship-engine data. Exploratory analysis revealed important patterns across the six features, while data cleansing ensured the dataset was reliable for modelling. Distribution analysis, pair plots, and box plots provided deeper insight into normal and abnormal behaviour. The IQR method offered a simple starting point but was insufficient for capturing complex interactions within the data. Isolation Forest proved to be a more effective approach, identifying multivariate anomalies that may indicate developing anomaly or unusual operating conditions.